CAUSALITY ASSESSMENT USING NARANJO SCALE AND WHO-UMC SCALE OF

CASE SCENARIO I HAVE SELECTED:

2. CAUSALITY ASSESSMENT

Naranjo Adverse Drug Reaction Probability

Question	Yes	No	Do Not Know	Score
1. Are there previous *conclusive* reports on this reaction?	+1	0	0	+1
2. Did the adverse event appear after the suspected drug was administered?	+2	-1	0	+2
3. Did the adverse reaction improve when the drug was discontinued or a *specific* antagonist was administered?	+1	0	0	+1
4. Did the adverse event reappear when the drug was re-administered?	+2	-1	0	0
5. Are there alternative causes (other than the drug) that could on their own have caused the reaction?	-1	+2	0	-1
6. Did the reaction reappear when a placebo was given?	-1	+1	0	0
7. Was the drug detected in blood (or other fluids) in concentrations known to be toxic?	+1	0	0	0
8. Was the reaction more severe when the dose was increased or less severe when the dose was decreased?	+1	0	0	0
9. Did the patient have a similar reaction to the same or similar drugs in *any* previous exposure?	+1	0	0	0
10. Was the adverse event confirmed by any objective evidence?	+1	0	0	+1
			TOTAL SCORE:	4

SCORING FOR NARANJO ALGORITHM: > 9 = DEFINITE ADR; 5-8 PROBABLE ADR; 1-4 = POSSIBLE ADR; 0= DOUBTFUL ADR

WHO – UMC causality categories

Causality term	Assessment criteria
Certain	• Event or laboratory test abnormality, with plausible time relationship to drug intake • Cannot be explained by disease or other drugs • Response to withdrawal plausible (pharmacologically, pathologically) • Event definitive pharmacologically or phenomenologically (i.e. an objective and specific medical disorder or a recognised pharmacological phenomenon) • Rechallenge satisfactory, if necessary
Probable / Likely	• Event or laboratory test abnormality, with reasonable time relationship to drug intake • Unlikely to be attributed to disease or other drugs • Response to withdrawal clinically reasonable • Rechallenge not required
Possible ✓	• Event or laboratory test abnormality, with reasonable time relationship to drug intake • Could also be explained by disease or other drugs • Information on drug withdrawal may be lacking or unclear
Unlikely	• Event or laboratory test abnormality, with a time to drug intake that makes a relationship improbable (but not impossible) • Disease or other drugs provide plausible explanations
Conditional/ unclassified	• Event or laboratory test abnormality • More data for proper assessment needed, or • Additional data under examination
Unassessable / Unclassifiable	• Report suggesting an adverse reaction • Cannot be judged because information is insufficient or contradictory • Data cannot be supplemented or verified

CASE SCENARIO : 1 USED TO FOR ASSESSING CAUSALITY IS

Patient name : S A

Age: 72 year

Gender: Male

Allergy status : No known allergy

Past Medical history : Hypertension, Type 2 Diabetic mellitus, Old CVA

Present illness : Recurrent ais - right mca-pca watershed infarct + right parietal infarct with left hemiplegia, old right ica distal occlusion (only on tab stiloz),dvt left ll, ?sepsis with septic shock, ckd, ild - usual interstitial pneumonia, type 2 dm (hba1c: 7.7%),systemic hypertension, hfref (ef: 34%), Anemia of chronic disease, old pulmonary tb (30 years back)

Past medication history : Tab Stiloz 50 1-0-1
Tab Atorva 0-0-1
Tab Amlodac AT (Amlodipine & Atenolol)
Tab Dytor 10 1-0-0
Tab Cospiaq Sm (Empa+Sita+MF) 25/100/100mg 1/2-0-1/2

A 72 Year old male patient was Started on 18/9/2026 Tab Bisobis (Bisoprolol) 2.5mg For HFrEF and Hypertension. Later on 19/9/2026 he developed Hypotension. The drug was stopped. On 20/09/2026 Blood pressure came to normal 130/76mmhg.

Laboratory data :

Physical examination: on 19/9/2026 Blood pressure: 90/60 mmhg ,

20/9/2026 Blood presuure:130/76mmhg

Management:

Tab Bisobis 2.5 mg was discontinued due to hypotension. Noradrenaline support was documented for hypotension. Blood pressure monitoring was advised every 2 hours.

On 20/09/2026 Blood pressure came to normal 130/76mmhg.

CASUALITY ASSESSMENT USING NARANJO SCALE AND WHO-UMC SCALE OF CASE SCENARIO OF THE TUTORIAL VEDIO

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5. Are there alternative causes (other than the drug) that could on their own have caused the reaction?	-1	+2	0	+2
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7. Was the drug detected in blood (or other fluids) in concentrations known to be toxic?	+1	0	0	0
8. Was the reaction more severe when the dose was increased or less severe when the dose was decreased?	+1	0	0	0
9. Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	+1	0	0	0
10. Was the adverse event confirmed by any objective evidence?	+1	0	0	0
		TOTAL SCORE:		6

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